## Cardiac Pathology: Chapter 5.2 R to L shunt




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| 1. **Read manga right to left**: right-to-left shunts (e.g. truncus arteriosus, TGV, tricuspid atresia, TOF, TAPVR) |
| 2. **Blue baby**: right-to-left shunts cause early onset cyanosis |
| 3. **Large purple trunk**: truncus arteriosus (right-to-left shunt) |
| 4. **Large V shirt**: truncus arteriosus often occurs with VSD |
| 5. **"September"** :failure of septation causes truncus arteriosus (no division between aorta and pulmonary trunk) |
| 6. **Neural crest shield**: failure of neural crest cell migration causes truncus arteriosus |
| 7. **"22" and "11"** helmet: failure of neural crest cell migration (e.g. truncus arteriosus) is associated with 22q11 deletion syndromes (e.g. DiGeorge) |
| 8. **Monster trainer station**: transposition of the great vessels (TGV - a right-to-left shunt) |
| 9. **Independent right and left circuits**: TGV results in an independent deoxygenated systemic circuit and oxygenated pulmonary circuit (aorta and pulmonary artery are transposed) |
| 10. **"PDA VS ASD":** TGV is incompatible with life without a VSD, ASD, or PDA (left-to-right shunts) |
| 11. **Murmur between the circuits**: TGV can present with a murmur form an associated left-to-right shunt |
| 12. **Red ponytail in front of blue**: with TGV, the aorta is anterior to the pulmonary artery (on cardiac echo) |
| 13. **Monster ball on a string** :with TGV, the cardiac silhouette looks like an "egg on a string" on CXR |
| 14. **Throwing up candy**: diabetes in the mother is a risk factor for TGV |
| 15. **Defeated spiral monster**: failure of the aorticopulmonary septum to spiral results in TGV |
| 16. **Tricuspid "Z" warrior**: tricuspid atresia (right-to-left shunt) |
| 17. **Hole in septal window**: tricuspid atresia is often associated with an ASD |
| 18. **"Tetra Sailors":** tetralogy of Fallot (Right-t-left shunt) |
| 19. **Earth sailor with constricted pulmonary trees**: pulmonary valve stenosis (feature of TOF) |
| 20. **Earth sailor's blue face**: pulmonary valve stenosis causes cyanosis (the degree of stenosis dictates the severity of disease) |
| 21. **Water sailor's conch shell**: TOF is associated with right ventricular hypertrophy (concentric) due to pressure overload |

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| 22. **Water sailor's large boots**: right ventricular hypertrophy (feature of TOF) seen on CXR as a "boot-shaped heart" |
| 23. **Fire sailor's big V neck**: ventricular septal defect (feature of TOF) |
| 24. **Air sailor's red ponytail flying overhead**: overriding aorta (feature of TOF) |
| 25. **Neural crest shield**: failure of neural crest cell migration causes TOF |
| 26. **"22" and "11" jewelry**: failure of neural crest cell migration (e.g. TOF) is associated with 22q11 deletion syndromes (e.g. DiGeorge) |
| 27. **Systolic spray**: TOF can present with a harsh systolic crescendo/decrescendo murmur (due to pulmonary valve stenosis) |
| 28. **Evil choking spell**: TOF can present with hypercapnic spells ("tet spells") |
| 29. **Squatting to dodge spell**: squatting relieves symptoms during a hypercapnic spell |
| 30. **Geyser shooting upward**: squatting increases SVR forcing more blood upward into the pulmonary circulation |
| 31. **"Tap VR":** total anomalous pulmonary venous return (TAPVR - right-to-left shunt) |
| 32. **Right, down, right, down...**: in TAPVR, the pulmonary veins drain back into the right heart (with the systemic circulation) |
| 33. **Dilated right tap dancer**: TAPVR causes a dilated right atrium and ventricle |
| 34. **Hole in septal window**: in TAPVR, an ASD allows some oxygenated to enter the systemic circulation |
| 35. **"Upstairs":** Ebstein's anomaly can present in infancy with cyanosis |
| 36. **Large atrium map**: Ebstein's anomaly is associated with dilation of the right atrium |
| 37. **"Event in atrium"**: the abnormal dilation of the right atrium and inferior displacement of the tricuspid valve into the ventricle istermed "atrialization" of the right ventricle |
| 38. **Regurgitation on tricuspid wig**: Ebstein's anomaly is associated with a malformed tricuspid valve and tricuspid regurg |
| 39. **Failing heart balloon**: Ebstein's anomaly is associated with right sided heart failure (due to severe tricuspid regurg) |
| 40. **Pregnant mother on "lift-ium":** Ebstein's anomaly is caused by lithium exposure in utero |